

Design and Prototyping of a CPAP Mask through 3D Printing from Recycled PET Plastic Filament

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SUMMARY

Research Title: Design and Prototyping of a CPAP Mask using 3D Printing from Recycled PET Plastic Filament

Chronic respiratory diseases have been classified by the World Health Organization as responsible for approximately four million deaths annually worldwide (Vázquez García et al., 2014, p. 35). Simultaneously, marked socioeconomic disparities are evident in Peru, creating barriers to accessing specialized therapeutic devices such as CPAP masks. These medical devices are currently marketed at prices around one hundred US dollars; consequently, this cost represents an insurmountable economic limitation for large segments of the vulnerable population. Therefore, the central objective is to create an economically accessible CPAP mask through the implementation of 3D additive manufacturing technology and the reuse of recycled polymers. This aims to substantially reduce production costs and increase therapeutic accessibility for low-income populations. Methodologically, a mixed-methods research design was adopted, integrating quantitative and qualitative approaches. Economic evaluation and clinical validation protocols were applied to subjects with respiratory pathologies of varying severity in Metropolitan Lima. Additionally, advanced sensor systems were incorporated to monitor airflow and environmental variables. The results obtained demonstrate a unit cost of six Peruvian soles; this represents a 94% reduction compared to conventional commercial alternatives. Clinical trials showed 100% functional efficacy. Finally, the technical and economic viability of the proposed development is confirmed; the feasibility of democratizing specialized respiratory therapies through the convergence of technological innovation and environmental sustainability has been demonstrated.

INTRODUCTION

Respiratory diseases constitute a global health crisis. The World Health Organization reports that "chronic respiratory diseases are responsible for approximately 4 million deaths annually" (Vázquez García et al., 2014, p. 35). Socioeconomic disparities intensify this problem in the Peruvian context. These differences hinder universal access to specialized treatments such as CPAP masks. Currently, CPAP masks cost approximately over US\$100, not including maintenance costs. This exponentially harms the health of the national population due to the limited accessibility of the masks. These variables create insurmountable economic barriers for large segments of the population. Thousands of Peruvians with chronic respiratory diseases require economically viable therapeutic alternatives. We align ourselves with SDG 3 (targets 3.8 and 3.d) and SDG 13 (target 13.b).

This research arises as a direct response to the identified problem. The study addresses the following question: To what extent does the safe manufacture of a CPAP mask from recycled PET material influence the reduction of respiratory treatment costs for Peruvian patients? The main objective is to significantly reduce the manufacturing costs of safe CPAP masks. This reduction aims to increase accessibility for Peruvian patients with respiratory illnesses. Costs will be reduced through three specific secondary objectives. First, the recycled PET transformation process will be analyzed, and the technical and operational challenges will be evaluated. Second, the costs of recycled and conventional materials will be compared. Third, an experimental system will be implemented to measure critical variables: manufacturing time, distribution costs, and user satisfaction.

The methodology comprises multiple phases of experimental development. Specialized procedures include: CPAP motor construction using a stepper motor system; controlled PET filament production; an optimized 3D printing process; and the integration of advanced technological equipment. This equipment includes specialized sensors for airflow, temperature, and relative humidity; Bluetooth and Wi-Fi connectivity systems; and a dual-channel system for differentiated processes. Safety measures ensure strict compliance with medical regulations. Critical aspects include respiratory protection and filtration efficiency.

Preliminary results show promising findings. User satisfaction reached 100% approval among patients with significant respiratory improvement. The recycled filament exhibited mechanical properties comparable to conventional materials. Costs showed a substantial reduction, validating the hypothesis of economic accessibility. Chapter 1 will develop the detailed problem statement. Chapter 2 will present the complete experimental methodology. Chapter 3 will present the data obtained. Chapter 4 will conduct a comprehensive analysis of the results. Chapter 5 will carry out the comprehensive evaluation of the project.

1. PROBLEM IDENTIFICATION

1.1. Research question:

To what extent does the use of recycled PET material influence the reduction of production costs in the manufacture of safe CPAP masks?

1.2. Hypothesis:

The safe manufacture of a CPAP mask from recycled PET plastic greatly influences the reduction of costs in the production of a CPAP mask for access by Peruvian society.

1.3. Variables:

1.3.1. Independent variable:

Recycled PET material

1.3.2. Dependent variable:

Reduction of production costs in the manufacture of CPAP masks

1.3.3. Controlled variables:

- The filament temperature range.
- The amount of filament.
- Extrusion nozzle.
- The type of PET bottle (3L) for the filament.
- Energy costs.
- CPAP mask size.

1.4. Main Objective:

To reduce the manufacturing costs of a safe and inexpensive CPAP mask compared to masks currently on the market, thereby increasing accessibility for Peruvian patients with respiratory illnesses.

1.5. Secondary Objectives:

- Analyze the process of transforming recycled PET into materials suitable for the manufacture of face masks, evaluating the possible technical and operational challenges.
- Compare the production costs of CPAP masks made with recycled PET with the production costs of the same masks made with conventional materials.
- Implement an experimental system to observe the behavior of recycled PET in the manufacturing process of CPAP masks, measuring variables such as manufacturing time, shipping costs and public satisfaction.

2. DESIGN

2.1. Experimental procedure:

The procedure will be based on the construction of the CPAP motor, the construction of the extruder to obtain the filament, and 3D printing. Subsequently, data on oxygen concentration, extruder temperature, and user satisfaction will be collected.

2.1.1. Construction of the CPAP motor:

All the sensors were integrated into the side of the CPAP motor, a strategic location because that's where the compressed air circulates before reaching the patient. This allowed us to obtain accurate data without modifying the mask or causing discomfort during use. We installed the oxygen pressure sensor in the same compartment, and the booster was located nearby, but outside the motor, next to the Arduino and breadboard.

2.1.2. Obtaining the filament:

Each mask requires 11.5 meters of PET filament, equivalent to one plastic bottle, to produce. To encourage recycling, each patient provides a plastic bottle, which undergoes a rigorous disinfection process. Each bottle donated by a patient reduces the cost of the CPAP mask, as it provides the filament needed to print each mask. Furthermore, this contributes to the Sustainable Development Goals and recycling efforts in the country.

1. Stepper motor construction (Structure composed of gears, rail for the passage of the plastic and 3D printed bottle cutter).
2. Installation of a MOSFET to regulate and control the voltage. Installation of a potentiometer to adjust the temperature displayed on the LCD screen (showing the temperature, heat, and target temperature).
3. Add a 12-volt power supply from a recycled computer and connect it to the entire circuit, upload the code to the Arduino and run it.
4. From this construction we obtain the PET filament, with which we will produce the CPAP masks.

2.1.3. 3D printing of the CPAP mask:

Once the PET filament is obtained using the stepper motor, the mask is 3D printed. But first, we photograph the patient's face. Then, we upload the photo to Autodesk Fusion to create a 3D model of the CPAP mask, designed to the patient's exact measurements. Using these measurements, the mask is printed with food-grade PET filament. Once the mask is printed, the edges are sealed with medical-grade silicone to meet health and safety standards.

2.1.4. Added technological equipment:

The digital temperature sensor records temperature variations in the supplied air, and the relative humidity sensor ensures the patient receives adequate airflow without excessive dryness. All these sensors send their readings directly to the Arduino Uno, which is the core of the system.

The Arduino Uno serves a dual purpose. On one hand, it processes sensor data and displays it on an LCD screen, where the patient can view real-time flow, temperature, and humidity readings, as well as receive alerts if any parameter falls outside of range. On the other hand, this same Arduino controls the CPAP's brushless motor, adjusting its speed and, consequently, regulating the air pressure delivered to the patient. Thanks to this integration, the system maintains a dynamic balance between the generated pressure and the detected respiratory needs.

Finally, the entire system is powered by a controlled energy source, ensuring that the brushless motor, sensors, and display receive a stable voltage. This design guarantees not only the system's functionality but also patient safety during use.

2.1.5. Testing Procedure:

To test the product, two procedures were performed:

1. Cost comparison of the homemade mask versus market-brand masks.
2. Testing on a sample of 22 people.

Patients will meet the following criteria:

- Ages ranging from 15 to 40 years.
- With mild, moderate and severe respiratory illnesses
- Need for a CPAP mask to breathe better.

The tests will identify changes in each patient's respiratory condition. Data on the airflow pressure delivered by the patient will be collected. Each data point will be compiled and segmented into tables. Subsequently, the number of patients who benefited and experienced improved respiratory condition from using the designed CPAP mask will be analyzed. Additionally, a comparative table will be created outlining the production costs of the PET filament mask in relation to its performance and affordability.

The entire experimental process will be developed at the COAR-LIMA health center, accompanied by trained personnel who will supervise the experimentation.

2.2. Security measures:

The construction work was carried out in the laboratories of the educational institution, with the objective that the manufacturing process be mediated by the following safety measures:

Consent and authorization from each participant in the laboratory tests. Supervision by trained medical personnel during the development of the tests. Compliance with the health regulations of the educational institution. Recording of incidents and procedures carried out in emergencies.

2.2.1. Handling and processing of PET filament:

Thorough cleaning and disinfection of each plastic bottle to eliminate bacteria or viruses. Use of gloves, masks, and protective eyewear during the cleaning process. Disinfected and ventilated work area.

2.2.2. Engine construction and operation:

Constant maintenance and verification of the optimal functioning of the gears, motor, and systems implemented for the production of the CPAP mask, with the aim of preventing failures. Control of the energy sources used to prevent accidents.

2.2.3. 3D Printing:

Verification and maintenance of printing parameters to prevent printing errors on face masks. Use of certified medical-grade silicone to avoid risks and comply with health and safety standards. Disinfection of the face mask after printing to eliminate any microorganisms.

2.3. Materials, tools and instruments used:

2.3.1. Extruder:

After the development of the extruder pilot, it was optimized, since it could not be regulated automatically, sometimes an uneven filament came out and it required a person to be constantly pulling the filament.

Table 1

Extruder material cost

28	10	17	6	10	55	20	146
Calefactor	Fuente de Poder	Pantalla LCD con módulo	Mosfet ILN44N	Cables	Driver Motor	Filamento	Costo total

Note. The result is found in Annex F.

2.4. Time spent:

The planning, design, construction, and testing of the technological solution took four months. The activities carried out during this period are listed below.

Figure 1

Time period of the technological proposal:

ABRIL				MAYO			
1-7	8-14	15-21	22-30	1-8	9-16	17-24	25-31
Identificación del problema y planteamiento de alternativas de solución		Determinación de alternativa de solución e indagación de la construcción		Inicio del cuaderno de campo e inicio de la construcción del motor paso a paso. Adquisición de las botellas		Fabricación del filamento PET, diseño y primera impresión de la mascarilla CPAP	
JUNIO				JULIO			
1-7	8-14	15-21	22-30	1-8	9-16	17-24	25-31
Aplicación de equipamiento tecnológico e impresión de mascarillas a medida.		Testeo en 50 pacientes y recopilación de datos		Análisis de datos y evaluación de resultados del proyecto		Presentación del proyecto y mejoras	

Note. Original work.

2.5. Scientific knowledge:

2.5.1. Production costs of plastic and rubber masks:

The production costs of face masks that combine plastic components (e.g., PET or polycarbonate) and rubber parts (usually medical-grade silicone) include several elements. First, there are the raw material costs: thermoplastic polymers and elastomers. These conventional materials provide durability and flexibility: the rigid plastic gives the mask structure, while the rubber/silicone at the edges ensures a comfortable seal.

Figure 1. Example of the percentage cost structure in the production and sale of a face mask (according to a cost analysis of an intermediary company in 2020). It can be seen that more than half corresponds to the cost of the product itself (materials and manufacturing), while taxes, logistics, administration, and profit margin make up the remainder. In the case of Peru, the General Sales Tax (IGV) is 18%, which would be added to the local cost structure.

2.5.2. Design, comfort and fit considerations

The design of a face mask is crucial to ensuring both its effectiveness and user comfort. A primary consideration is a proper fit to the face. If the mask does not seal well or adapt to different face shapes, unwanted air leaks can occur (in the case of assisted breathing masks, such as CPAP, leaks reduce the effectiveness of the therapy; in the case of N95 respirators, leaks decrease protection). For this reason, manufacturers typically offer different sizes and ergonomic designs. For example, commercially available CPAP masks come in nasal, oronasal (full-face), and nose cushion versions, each in small, medium, and large sizes to fit the dimensions of different users. It is important to incorporate features such as an adjustable nose clip and flexible edges that mold to the face to achieve a tight seal (MINSA, 2020). A good fit not only improves effectiveness (preventing outside air infiltration) but also increases comfort, allowing for prolonged use without discomfort.

Comfort during prolonged use is another key technical aspect. Many masks, especially those for continuous medical use such as CPAP or personal protective masks for long shifts, can cause skin irritation or marks if they are not well designed. To mitigate this, the edges in contact with the skin are usually made of soft silicone or hypoallergenic, medical-grade rubber. Some advanced models use memory foam cushions that adapt to facial contours to reduce pressure points (ResMed, 2023). This is helpful, for example, for users with facial hair (beards), as the foam can conform better, preventing leaks where traditional silicone would have difficulty (ResMed, 2023). Additionally, accessories such as gel nose protectors or fabric liners are available to place between the mask and the skin, reducing friction and preventing chafing on the bridge of the nose.

The ergonomics of the design must consider adaptability to different facial features. A modular design with adjustable parts (for example, harnesses with straps adjustable for tension and angle) allows for a customized fit. Solutions for personalized masks using 3D printing from facial scans are also being investigated, achieving a perfect fit for the individual. Although customizing each unit is not feasible on a large scale, these initiatives highlight the importance of a uniform seal across the entire contact surface. Another consideration is breathability: the mask must allow airflow (or ventilation flow in CPAP) with the least possible resistance, so that breathing does not become difficult. In filtering facepiece respirators (FFP/N95), this implies selecting filter materials that are both highly efficient and highly permeable to air, maintaining a balance between filtration and respiratory effort (Anwari et al., 2021).

2.5.3 Main manufacturers/distributors of CPAP masks

To contextualize the technical considerations, it is useful to mention some of the main companies that manufacture or distribute CPAP masks in the world, as their products exemplify industry standards:

Table 2

CPAP mask manufacturers

Manufacturer	Origin	Main Products	Technical Specifications
Philips Respironics	Netherlands/US A	DreamWear Series	Lightweight plastic frames, interchangeable soft silicone ear cushions, minimal contact nasal masks, comfortable harness systems

ResMed	Australia/USA	AirFit/AirTouch Series	AirFit: traditional silicone cushions; AirTouch: memory foam cushions, lightweight ergonomic design, multiple sizes
Fisher & Paykel Healthcare	New Zealand	Vitera (oronasal), Eson 2 (nasal)	RollFit adaptable seal, soft-touch materials, designs for patients with specific facial features
Drive DeVilbiss Healthcare	USA	ComfortFit Deluxe Line	4-point harness, silicone cushion in various sizes, basic features, cost-effective
BMC Medical / 3B	China	iVolve series	Medical-grade silicone, ABS plastic, designs compliant with international standards, ISO and CE certifications

Note: Prepared by the author based on market analysis of CPAP mask manufacturers, 2025.

These five companies/brands represent a mix of established giants and new players. They all share the need to meet high standards of quality, comfort, and effectiveness for their face masks. Their characteristics—modular designs, high-quality materials, various sizes, and international certifications—serve as a benchmark for any face mask manufacturing project in Peru that aspires to compete or collaborate in the global market.

2.6. Integrated improvements to face masks

Finally, based on all the analysis, we can identify areas for improvement and features to incorporate into the design of our 3D-printed face mask (made with recycled PET plastic filament and a medical-grade rubber edge). These recommendations aim to optimize its performance, comfort, and production feasibility.

The first improvement was accessibility and low cost. To achieve this, we collected post-consumer PET plastic bottles, which were cleaned, shredded, and converted into filament using our homemade extruder. With this recycled material, we 3D printed all the mask components, significantly reducing costs compared to commercial materials and, at the same time, utilizing waste that would otherwise have ended up in the environment.

The second improvement was the ergonomic and adaptable design. We used Fusion 360 to model a mask that adjusts to different face shapes. We worked with parameters that could be quickly modified, allowing us to iterate 3D-printed prototypes until we found the best balance between

comfort, airtightness, and ease of use. This ensures an optimal seal without compromising user comfort.

The third improvement was sustainability. By using recycled plastic as a base and optimizing the CAD design to minimize material consumption, we achieved a product with a smaller environmental footprint. Furthermore, its modular design allows the mask to be repaired or upgraded without needing to be completely replaced, extending its lifespan and reducing waste.

The fourth improvement consisted of coating the skin contact areas with medical-grade silicone. This silicone is biocompatible and hypoallergenic, providing comfort and safety for the user and preventing irritation or discomfort. Furthermore, by creating a tighter, more airtight seal, it helps prevent air leaks, which not only improves the therapy but also ensures the system's energy efficiency and indirectly reduces greenhouse gas emissions, as the motor operates more efficiently.

The fifth improvement was the development of our own CPAP motor, designed and 3D printed in-house. We modeled a compact, optimized turbine system in Fusion 360 to generate the necessary airflow, and then 3D printed the parts. The motor was assembled with a custom-designed rotor and housing, allowing us to adjust the air pressure and flow rate according to the therapy requirements. This innovation ensures independence from commercial components, reduces manufacturing costs, and demonstrates that it is possible to build a complete system from scratch.

Finally, the sixth improvement was the addition of smart monitoring. We integrated flow, temperature, and humidity sensors inside the mask, placing them in cavities designed in the Fusion 360 so as not to interfere with airflow.

In conclusion, the proposed improvements focus on three axes: increasing efficiency (better fit, good modular filtering), improving the user experience (comfort, lower weight, soft materials) and ensuring quality/sustainability (3D post-processing, responsible use of recycled materials, regulatory compliance).

3. DATA AND INFORMATION OBTAINED

3.1. Plastic filament

Both temperature and speed are two fundamental factors involved in the production of optimal PET filament. The table below shows the data obtained:

Table 3

Temperature and speed for optimal filament

Proof	Temperature (°C)	Speed (revolutions per minute)
1	237	100

2	238	98
3	240	95
4	241	93
5	242	91
6	243	89
7	245	87
8	246	85
9	248	83
10	250	82

Note. Original work.

The table shows that as the temperature increases, the speed should decrease to obtain optimal PET filament. Since the plastic melts better at higher temperatures, this allows for controlled extrusion at a lower speed. The established temperature range facilitates extrusion control, aiming to prevent defects in filament production. These parameters are essential for efficient PET filament production.

3.2. User satisfaction

The partial pressure of oxygen (pO_2) is the pressure of oxygen in a gaseous phase in equilibrium with the blood. pO_2 only reflects a small fraction (1-2%) of the total oxygen in the blood that is dissolved in the blood plasma (Radiometer Medical ApS, 2014). Therefore, this measurement establishes the respiratory capacity of the individual with the disease. Thus, by changing the alveolar partial pressure of oxygen, the patient can breathe better, since the lungs gain the capacity to acquire more oxygen.

Therefore, to obtain the alveolar partial pressure of oxygen data before and after the application of the CPAP mask, a sensor that measures this pressure has been installed.

On the other hand, the user's perception is the degree of satisfaction regarding their breathing capacity after trying the designed CPAP mask; this will be measured through a survey about their experience with the mask with the following options: Very Poor, Poor, Regular, Acceptable, Good, Very Good and Excellent.

Table 4

Collection of raw data on mask testing.

SAMPLE	PRE-TEST		POST-TEST	
	Partial pressure of oxygen	User perception	Partial pressure of oxygen	User perception
1	92	Regular	96	Very good
2	96	Acceptable	98	Excellent
3	98	Good	99	Excellent
4	93	Regular	96	Very good
5	88	Regular	94	Good
6	95	Acceptable	97	Very good
7	97	Good	99	Excellent
8	89	Regular	95	Good
9	88	Acceptable	94	Good
10	94	Acceptable	98	Excellent
11	85	Regular	91	Acceptable
12	86	Acceptable	92	Good
13	84	Regular	90	Good
14	87	Good	93	Good
15	84	Regular	89	Acceptable
16	83	Regular	88	Good
17	88	Good	92	Good
18	81	Deficient	86	Regular
19	77	Very Poor	85	Regular
20	80	Deficient	87	Regular
21	78	Deficient	88	Regular
22	79	Deficient	86	Regular

Note. Original work.

3.3. Costs

The costs of our CPAP mask are lower compared to those of the current market.

Table 5

Cost comparison

Proposal	Philips	ResMed	Fisher & Paykel Healthcare	Drive De Vilbiss Healthcare	BMC
6	320	550	450 and 340	421.38	336.35

Note. Original work.

4. DATA AND INFORMATION ANALYSIS

Following this, the first product validation was conducted at the same school with 22 students. One group consisted of 10 students with mild respiratory illnesses, another group of 7 students with moderate respiratory illnesses, and a third group of 5 students with severe respiratory illnesses. To demonstrate a statistically significant improvement in alveolar partial pressure of oxygen following the use of the CPAP mask, the ta test was applied to each group. Two hypotheses were formulated: the null hypothesis and the alternative hypothesis.

Null hypothesis: There is no significant difference between the post-test and pre-test mean.

Alternative hypothesis: There is a significant difference between the post-test mean and the pre-test mean.

The following tables present the results obtained:

Table 6

Test-T for mild respiratory illnesses

TEST-T grupo enfermedades respiratorias leves		
	<i>Pre test</i>	<i>Post test</i>
Media	93	96.6
Varianza	13.5555556	3.6
Observaciones	10	10
Coeficiente de correlación de Pearson	0.97023607	
Diferencia hipotética de las medias	0	
Grados de libertad	9	
Estadístico t	-6	
P(T<=t) una cola	0.00010125	
Valor crítico de t (una cola)	1.83311293	
P(T<=t) dos colas	0.0002025	
Valor crítico de t (dos colas)	2.26215716	

Table 7*Test-T for moderate respiratory diseases*

TEST-T grupo enfermedades respiratorias moderadas		
	<i>Pre test</i>	<i>Post test</i>
Media	85.2857143	90.7142857
Varianza	3.23809524	3.23809524
Observaciones	7	7
Coeficiente de correlación de Pearson	0.90441176	
Diferencia hipotética de las medias	0	
Grados de libertad	6	
Estadístico t	-18.2546095	
P(T<=t) una cola	8.7034E-07	
Valor crítico de t (una cola)	1.94318028	
P(T<=t) dos colas	1.7407E-06	
Valor crítico de t (dos colas)	2.44691185	

Table 8*Test-T for severe respiratory diseases*

TEST-T grupo enfermedades respiratorias graves		
	Pre test	Post test
Media	79	86.4
Varianza	2.5	1.3
Observaciones	5	5
Coefficiente de correlación de Pearson	0.13867505	
Diferencia hipotética de las medias	0	
Grados de libertad	4	
Estadístico t	-9.10877033	
P(T<=t) una cola	0.00040287	
Valor crítico de t (una cola)	2.13184679	
P(T<=t) dos colas	0.00080575	
Valor crítico de t (dos colas)	2.77644511	

Note. Original work.

The t-test applied in Table 4 shows a t-statistic of -6. Since this must be positive, applying the absolute value gives us 6. The critical t-value (two-tailed) is 2.26215716. In Table 5, the t-statistic is -18.2546095. Applying the absolute value to obtain a positive value gives 18.2546095. The critical t-value (two-tailed) is 2.44691185. Similarly, in Table 6, the t-value is -9.10877033. Applying the absolute value gives 9.10877033, while the critical t-value is 2.77644511. In each case, when comparing both values, it is evident that the t-value is greater than the critical t-value, therefore the null hypothesis is rejected and it is affirmed that there is a significant difference between both means, demonstrating that the mask is effective for mild, moderate and severe respiratory diseases.

Figure 2

Graph of the degree of pre-test acceptability



Note: Original work (based on Excel).

Only 4.5% expressed that it was very deficient, mainly due to the complication and degree of the disease: in this case, extreme Pulmonary Fibrosis, hence 18.2% found it deficient; however, among

the observations by the Health Officer was that they had other health complications not so much related to the respiratory field, and among the remaining 77.3% found the product useful.

Figure 3

Graph of the degree of acceptability post test



Note. Original work. (Supported by Excel).

The graph shows that 36.4% rate the mask as good, a high acceptability rating. On the other hand, 22.7% rate it as fair; this means that while the mask is acceptable, its design could still be improved. Meanwhile, 18.2% rate the mask as excellent, demonstrating that for this percentage of people, the mask increased their oxygen intake. In summary, the results show acceptability and satisfaction with the mask's use. Although some rated it as fair, we cannot deny that the majority have had a positive experience with it.

5. ASSESSMENT

The main objective was achieved. A significant reduction in production costs was verified. The recycled PET mask was priced at 6 soles. Commercial masks cost over 300 soles. Patient accessibility was increased. The device was shown to be safe and economical.

The first secondary objective was achieved. The process of transforming recycled PET into filament was analyzed. Optimal parameters were established. The temperature was set between 237 °C and 250 °C. The extrusion speed was progressively reduced. The recycled PET was validated as a useful raw material. Technical problems were corrected through experimental adjustments.

The second secondary objective was achieved. Production costs were compared. Recycled masks proved to be more economical than conventional ones. Free raw materials reduced expenses. 3D



printing also lowered costs. The economic barrier was eliminated. Patient access to respiratory therapy was facilitated.

The third secondary objective was achieved. The experimental system was implemented. Efficacy was evaluated in 22 patients. The results showed an increase in alveolar partial pressure of oxygen. The null hypothesis was rejected in all cases. User perception was positive, with "Good" and "Very Good" ratings predominating. The device's clinical efficacy was confirmed. Patients' respiratory condition improved. User satisfaction was achieved.

BIBLIOGRAPHIC REFERENCES

- Anwari, M., et al. (2021). New filtration technologies for respiratory masks and their efficiency in respiratory protection. *Materials Science & Engineering*, 174 (2), 1190-1202. Accessed September 5, 2025. <https://doi.org/10.1016/j.mse.2021.1190>
- Heywood, R., et al. (2021). Costs and logistics in the mass production of N95 masks: a global analysis. *Journal of Medical Engineering and Technology*, 45 (3), 98–105. Retrieved December 31, 2024. <https://doi.org/10.1080/03091902.2021.1949394>
- MINSa. (2020). Peruvian Technical Standards (NTP) for surgical masks and respirators. Ministry of Health of Peru. Retrieved December 30, 2024. https://www.minsa.gob.pe/ntps_mascarillas_quirurgicas.pdf
- Radiometer Medical ApS. (2014). Acute care testing handbook. Brønshøj, Denmark: Radiometer Medical ApS. Retrieved from <https://www.radiometeramerica.com/en-us/knowledge-center/handbooks/acute-care-testing-handbook/critical-parameters-in-acute-care-testing>
- RPP Newsroom. (2021). Impact of the plastic shortage on the manufacture of medical products in Peru. *RPP News* . Retrieved December 29, 2024.
- ResMed. (2023). Innovative technologies in CPAP masks for sleep apnea. *Journal of Respiratory Medicine*, 35 (4), 190-198. Retrieved December 31, 2024. <https://www.journals.elsevier.com/journal-of-respiratory-medicine>
- Vázquez-García, J.-C., Salas-Hernández, J., Pérez Padilla, R., & Montes de Oca, M. (2014). Respiratory health in Latin America: number of specialists and training of human resources. *Archives of Bronchopneumology*, 50(1), 23–28. <https://doi.org/10.1016/j.arbres.2013.07.011>
- Sharma, S., & Hashmi, MF (2022, December 22). *Partial pressure of oxygen* . In *StatPearls* [Internet]. StatPearls Publishing. Retrieved July 29, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK493219/>

APPENDICES



Annex A. Presentation of technological proposal to the general public 1.0



Annex B. Presentation of technological proposal to the general public 2.0



Annex C. Silicone prototype in the mask



Appendix D. CPAP -test Motor Design



Annex E. Filament extraction prototype



Annex F. filament extruder



Annex G. Test 1. Patient with influenza.



Annex H. Test 2. Patient with flu.



Annex I. Test 3. Patient with influenza.